



KIET
DEEMED TO BE
UNIVERSITY
DELHI - NCR, INDIA
(Under Section 3 of the UGC Act, 1956)



A Project Report

on

NourishShare: A Real-Time Geospatial Matching Platform for Urban Food Redistribution

submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY DEGREE

SESSION 2025-26

in

COMPUTER SCIENCE & ENGINEERING (AI & ML)

By

Adarsh Gupta (2200291520011)

Anushka Singh Yadav (2200291520043)

Archi (2200291520045)

Under the supervision of

Dr. Pramod Kumar Sethy

Dr. Rekha Kashyap

KIET Group of Institutions, Ghaziabad

Affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)

May 2026

DECLARATION

We hereby declare that the project report entitled “**NourishShare: A Real-Time Geospatial Matching Platform for Urban Food Redistribution**” submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering (AI)** is our original work.

This work has not been submitted previously for the award of any degree or diploma in any university or institution. All sources of information used in this project have been properly acknowledged.

Adarsh Gupta (2200291520011)

Anushka Singh Yadav (2200291520043)

Archi (2200291520045)

Date:



KIET
DEEMED TO BE
UNIVERSITY
DELHI - NCR, INDIA
(Under Section 3 of the UGC Act, 1956)

CERTIFICATE

This is to certify that the project report titled **“NourishShare: A Real-Time Geospatial Matching Platform for Urban Food Redistribution”** submitted by **Adarsh Gupta, Anushka Singh Yadav, and Archi** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering (AI)** from **Dr. A.P.J. Abdul Kalam Technical University, Lucknow**, is a record of the work carried out by them under my supervision.

The work embodied in this report is original and has not been submitted for the award of any other degree or diploma.

Supervisor Name

Pramod Kumar Sethy

(Designation)

Assistant Professor

CSE (AI)

Date:

ACKNOWLEDGEMENT

We would like to express our sincere gratitude for the successful completion of this project titled “**NourishShare: A Real-Time Geospatial Matching Platform for Urban Food Redistribution.**”

The authors express sincere gratitude to **Mr. Pramod Kumar Sethy**, Assistant Professor, Department of Computer Science & Engineering (AI), KIET Group of Institutions, for his valuable guidance, encouragement, and continuous support throughout the development of this project.

Gratitude is extended to the faculty members of the Department of CSE (AI) for providing the necessary resources and support during the course of this work.

Finally, special thanks are due to our family members and friends for their encouragement, motivation, and constant support throughout the completion of this project.

Date:

Adarsh Gupta (2200291520011)

Anushka Singh (2200291520043)

Archi (2200291520045)

ABSTRACT

Urban areas often have a problem with food waste and food security. Every day a substantial volume of food is wasted by households, restaurants, and event planners, while many parts of society lack ready access to food. The absence of formal coordination between food providers and non-governmental organizations (NGOs) can result in inefficient redistribution of food.

This project introduces NourishShare, a real-time geospatial matching system that aims to redistribute surplus food. This method uses geospatial intelligence and intelligent matching to connect food donors with local non-governmental organizations (NGOs) through an interactive web platform.

For geocoding and real-time mapping, the system makes use of the Google Maps API and the MERN stack (MongoDB, Express.js, React.js, Node.js). NGOs can look for and claim local food donations based on geography, and food givers can make lists for excess items. To provide security, it employs JSON Web Tokens (JWT) for authentication.

By rapidly linking potential beneficiaries within a certain geographic area, the planned system will reduce food waste, optimize food allocation, and shorten coordination times. Scalability and future system enhancements, such as donation forecasting and route planning, are made possible by the system's modular design.

This research shows how scalable and effective urban food donation may be achieved by integrating new web and geospatial technology.

TABLE OF CONTENTS

Page No.

DECLARATION.....	ii
CERTIFICATE.....	iii
ACKNOWLEDGEMENT.....	iv
ABSTRACT.....	v
LIST OF FIGURES.....	viii
LIST OF TABLES.....	ix
LIST OF ABBREVIATIONS.....	x
 CHAPTER 1: INTRODUCTION	 1
1.1 Introduction.....	1
1.2 Problem Statement.....	2
1.3 Objectives	2
1.4 Scope of the Project.....	3
1.5 Organization of the Report.....	3
 CHAPTER 2: LITERATURE REVIEW.....	 4
2.1 Existing Food Redistribution Systems.....	4
2.2 Geospatial Matching Techniques.....	5
2.3 Digital Platforms for Social Resource Distribution.....	5
2.4 Limitations of Existing Systems.....	6
2.5 Research Gap.....	6
 CHAPTER 3: PROPOSED METHODOLOGY	 7
3.1 System Overview	7

3.2 System Architecture	7
3.3 Technology Stack.....	9
3.4 Data Flow of The System	9
3.5 Matching Methodology.....	10
3.6 Database Schema Design.....	10
 CHAPTER 4: IMPLEMENTATION AND RESULTS.....	 13
4.1 Implementation Details	13
4.2 Experimental Setup	15
4.3 Performance Evaluation.....	16
 CHAPTER 5: CONCLUSIONS AND FUTURE SCOPE.....	 17
5.1 Conclusion	17
5.2 Future Scope	18
 REFERENCES.....	 19
APPENDIX 1.....	21
 PATENT	
PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION	
PLAGIARISM REPORT	

LIST OF FIGURES

Figure No.	Description	Page No.
3.1	System Architecture of NourishShare Platform	8
4.1	Data Flow of Food Donation Matching Process	14
4.2	Donor Dashboard Interface	15
4.3	Food Donation Submission Form	16
4.4	Live Food Donation Map	17
4.5	NGO Claim Interface	18

LIST OF TABLES

Table No.	Description	Page No.
3.1	Technology Stack Used in the System	9
3.2	Database Schema for Food Redistribution Platform	10
3.3	API Endpoints of the System	10
3.4	Performance Evaluation Metrics	11

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
GPS	Global Positioning System
JWT	JSON Web Token
ML	Machine Learning
MERN	MongoDB, Express.js, React.js, Node.js
NGO	Non-Governmental Organization

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Food waste and food insecurity are two major global challenges that coexist in many urban environments. Large amounts of edible food are discarded every day by restaurants, households, catering services, and event organizers. At the same time, a significant portion of the population struggles to access sufficient food. The mismatch between food surplus and food demand highlights the need for efficient food redistribution systems. According to global reports, approximately one-third of all food produced is wasted annually, highlighting the urgency of efficient redistribution systems.

Traditional food donation mechanisms often rely on manual coordination through phone calls, social media messages, or volunteer networks. These approaches are unstructured and inefficient, resulting in delays in collecting surplus food and increased chances of food spoilage. Without a centralized digital platform, it becomes difficult to track available donations and identify suitable recipients quickly.

With the development of web and geographic technologies, there is a possibility of making digital tools that can be used to facilitate the distribution of excess food. It is expected that with such a process, the time required for the redistribution of food will become shorter as it will enable efficient communication between food donors and recipients.

For this reason, NourishShare will be proposed as a real-time geospatial food redistribution tool that connects food donors with NGOs located close to each other. The system relies on geospatial and web technologies to automatically match recipients and donors through the use of the MERN stack.

The system is built with the MERN stack, offering a scalable approach to support multiple users. The system uses the Google Maps API to enable NGOs to see food donations on a map and reserve them based on location and availability.

This system aims to address food waste and coordination inefficiency and build a robust digital platform for urban food redistribution.

1.2 PROBLEM STATEMENT

Surplus food is produced in abundance in urban areas, but it is often wasted because of poor coordination between donors and recipients. Current food donation systems typically rely on either manual communication tools, which are slow, unstructured, and inefficient, or digital platforms.

Donors often fail to find local organizations to pick up their food. NGOs also do not have timely information about the availability of food in their area. As a result, redistribution opportunities are missed and food goes to waste.

So there is a need for an online system that can effectively match food donors with local NGOs using real-time geospatial matching and automated coordination processes.

1.3 OBJECTIVES OF THE PROJECT

The main goals of the NourishShare framework are to design a highly effective digital system for food redistribution and to create a smooth interaction between the food donors and the closest NGOs by geospatial matching. In addition, the goal is to reduce the waste of food due to reduced delay in the process of picking up donations and to visualize the food availability through an interactive map system. Moreover, the framework will be scalable for multiple users at one time.

1.4 SCOPE OF THE PROJECT

The scope of the current project involves the design of a web portal that will allow the donors to post the details of the extra food products and help the NGOs in locating and claiming the donations using the interactive maps. By using spatial filtering and algorithmic matching techniques, the project aims to improve the efficiency of food delivery in urban areas.

In addition to supporting numerous concurrent users, the system has a security feature that uses token-based authentication to verify user identity. While the system shows promise for large-scale food redistribution, the current version operates in a controlled setting and uses simulated data for testing, primarily focusing on urban regions where food waste and demand overlap.

Additional features like donation prediction, improved collection routes, and mobile app

deployment might be included in future iterations.

1.5 ORGANIZATION OF THE PROJECT

The report contains five chapters. In Chapter 1, background, problem statement, objective, and scope of the project will be presented. In Chapter 2, we present a survey of the available systems and literature. Chapter 3 elaborates on the methodology that has been adopted for developing our system, which includes a description of its architecture and data flow.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Food redistribution mechanisms have attracted more and more attention over the past years as a reaction to the growing problems of food wastage and food insecurity. Thanks to the emergence of various digital technologies and geospatial tools, a number of platforms as well as research initiatives have been conducted with the aim of facilitating coordination among food suppliers and organizations providing charity assistance.

This chapter provides an analysis of previous literature and work regarding food redistribution mechanisms, geospatial coordination systems, and digital allocation systems. Some research and platforms have attempted addressing problems with food redistribution through digital and geospatial technologies.

2.2 EXISTING FOOD REDISTRIBUTION SYSTEMS

Some of the digital technologies used in addressing the challenge of food waste involve creating platforms that link donors to organizations that distribute the donated food among the needy members. Such platforms usually allow the donors, who are mostly restaurants, grocery stores, or individuals, to place their surplus food donations, which are collected by NGOs or food banks.

The use of technology in food donation has proved that digital coordination results in much better recovery of food products than the conventional process of coordinating food donations manually. Digital platforms enhance the efficiency of food distribution since they provide a central point for viewing all donations and therefore minimize the amount of time taken during communication.

But most of the current digital platforms depend on manual coordination between the donors and the recipients. The lack of proximity algorithms and optimization makes it difficult for these technologies to achieve real-time efficiency.

Therefore, while machine learning enhances adaptability and detection capacity, it has to be integrated with other methods to solve these problems.

Platforms like OLIO and Feeding India have demonstrated the effectiveness of digital coordination in reducing food waste even by connecting donors with beneficiaries.

2.3 GEOSPATIAL MATCHING TECHNIQUES

Geospatial technologies are very important in current logistics and resource management frameworks. Location-based technologies allow systems to locate users close to them, find optimal paths, and increase efficiency in delivery and distribution of resources.

Studies on geospatial matching have found that allocating resources by considering geographical proximity results in significant savings in transport time and cost. Location-based applications such as Uber, taxi-hailing apps, delivery applications, and logistics applications use spatial indexing and distance ranking to allocate service providers to clients.

These techniques can be adopted in food redistribution applications as well. Using these techniques, one could match NGOs closest to food donation and ensure that they pick up food in the shortest time to prevent spoilage.

The use of geospatial technologies is not common among most of the existing food redistribution applications. Methods such as spatial indexing and the Haversine formula are often used to calculate distances between geographical coordinates. Techniques such as spatial indexing and the Haversine formula are commonly used to calculate the distance between geographic coordinates in real-time systems.

2.4 DIGITAL PLATFORMS FOR SOCIAL RESOURCE DISTRIBUTION

In addition to the food redistribution process, digital technology has also been employed in other areas of social resource distribution, including health care services provision, disaster relief coordination, and volunteers' administration.

These systems' instantaneous information sharing and automated action coordination are essential for increasing the effectiveness and speed of resource allocation. Web applications and cloud computing technology enable coordinated actions among several participants. To guarantee safe user engagement, these platforms also frequently incorporate security and authentication methods including role-based access controls.

Within these platforms, users with various roles—such as administrators, providers, and recipients—interact with each other in a structured way. The NourishShare platform has embraced these technology concepts through modular web application design and role-based authentication. e.g., disaster relief apps, Uber Logic, etc.

2.5 LIMITATIONS OF EXISTING SYSTEMS

The advantages of digital platforms for resource coordination have been demonstrated by existing systems, however there are still some significant drawbacks. Since many current systems rely on manual coordination, which causes delays and inefficiencies, one such problem is the absence of automated matching algorithms that take geographic proximity into account.

Moreover, the absence of date-based matching greatly impacts the redistribution of food with limited shelf life, leading to the wastage of food due to time-insensitive matching. Lack of scalability is another issue, with many platforms unable to effectively manage a growing user base while ensuring minimal response time and optimal system efficiency.

Moreover, the lack of robust secure authentication mechanisms leaves these systems vulnerable to security issues such as data integrity and data security.

These challenges point to a need for a smarter, scalable, and secure food redistribution system.

2.6 RESEARCH GAP

However, based on an analysis of the available literature, various research gaps have been found.

There exist numerous food redistribution frameworks that have not developed decision-making mechanisms that are systematic. Geospatial matching and food expiration date-based allocation are common missing aspects of existing software frameworks.

Moreover, integration of scalable web architecture that allows for real-time donor and NGO collaboration has not been provided in any existing framework. Lack of such frameworks hampers efficient food redistribution operations and makes the process prone to waste.

In this regard, there is a need for development of a platform that will incorporate geospatial intelligence, automation of allocation mechanisms, and scalable web architecture in food redistribution systems.

Our proposed NourishShare system seeks to tackle these issues by incorporating proximity-based matching, real-time geospatial services, and secure authentication in a modular web application. This need identifies the lack of an integrated system that incorporates real-time geospatial matching, expiry-time matching, and scalable design in a unified platform.

CHAPTER 3

PROPOSED SYSTEM AND METHODOLOGY

3.1 SYSTEM OVERVIEW

The proposed NourishShare system is a real-time online system to support coordination between food donors and non-governmental organizations (NGOs). The system aims to minimize food waste through effective redistribution of surplus food through geospatial matching.

It matches donors (restaurants, event planners, grocery retailers, and households) with the closest NGOs that can pick up and deliver the food. Donors are able to submit information about the surplus food, such as food type, amount, location, and time of expiry.

When a donation is made, the system matches the details with appropriate NGOs within a specified geographic area. NGOs can then search for and claim the donated items in a map-based user interface based on availability and distance.

The system adopts a scalable web architecture and supports real-time user interactions. The system also provides secure access control using token authentication. The system uses a client-server architecture with real-time communication between the client and server for interactive data exchange.

3.2 SYSTEM ARCHITECTURE

The NourishShare platform is structured in a layered architecture, with the frontend layer, backend API layer, database layer, authentication layer, and integration with external services. The frontend is built with React.js to offer a dynamic user interface for donors and NGOs. The backend layer, which is constructed using Node.js and Express.js, controls API requests, application logic, and database communication.

MongoDB Atlas supports the database layer, which stores and retrieves donation and user data. JSON Web Tokens (JWT) are used to implement user authentication. Additionally, geospatial capabilities like mapping and geocoding are supported by integrating external services like the Google Maps API. Inter-component communication, scalability, and maintainability are all improved by this architecture.

Frontend Layer

The NourishShare system's frontend layer offers the graphical user interface that facilitates efficient user interaction with the platform and each other. React.js technology, which is renowned for its capacity to generate dynamic elements and create responsive designs, was used in its construction. Features like donor registration and login, entering information regarding surplus food, displaying locations for donations on a map, gaining access to the NGO dashboard for accepting donations, and offering real-time updates on food availability will all be available through the frontend. REST API requests will be used by the frontend to interact with the backend services.

Backend API Layer

Node.js and Express.js, which handle database connectivity, request management, and application functionality, are used to build the backend API. It manages critical tasks such user authorization and authentication, uploading food donations, geocoding physical addresses to latitude-longitude values, allocating donors to NGOs based on proximity, and handling donation claim requests. To ensure that every incoming request is authorized and has valid arguments, middleware validations are used.

Database Layer

MongoDB Atlas, a NoSQL database offered as a cloud service, is used by the database layer to efficiently store and manage all of the system's data. The database is divided into several primary collections, such as the Users collection, which contains information about users and their roles; the Food Posts collection, which keeps track of food type, quantity, locations, and expiration dates; and the NGO Claims collection, which records all donation claims and allocations.

Authentication Layer

The authentication layer provides security for the system via the use of JWTs. In this case, it ensures that only authenticated users have access to the functionality of the system. Once the

user has logged into the system successfully, a token will be generated and placed in each successive request to the API.

External Services Layer

External services have been included as well, especially those from the Google Maps API, which provides geospatial capabilities. This includes the usage of the Geocoding API, which converts text-based addresses into geographic coordinates. On the other hand, the Maps JavaScript API makes it possible to display food donation sites on a map. Another API, the Directions API, will be considered for the future.

3.3 TECHNOLOGY STACK

NourishShare is built with the MERN stack, comprising MongoDB, Express.js, React.js, and Node.js. MongoDB provides scalable and dynamic NoSQL database operations to manage user accounts, donation posts, and claims. Express.js handles server routing and API calls, and Node.js is the platform for running the backend server. React.js is used to create a dynamic user interface.

Other technologies used include Google Maps API for location services and mapping, JSON Web Tokens for secure user authentication, Axios for seamless API calls between the frontend and backend, and Tailwind CSS for responsive and modern user interface design. This stack enables scalable web application design and rapid development.

3.4 DATA FLOW OF THE SYSTEM

The initial step in the NourishShare system's information flow is donor registration and authentication. After successful authentication, the donor generates an information record containing details about the location, quantity, and time of expiry of the donated food item. The next step is geocoding of the address submitted by the donor using the Google Geocoding API, which converts the address into geographic coordinates. Next comes storing these details in the MongoDB database. Then begins the matching process wherein the system finds all NGOs present in a specified geographic range of the donor using a geofiltering technique. The eligible NGOs use the map interface to view the donations that are available for claim.

3.5 MATCHING METHODOLOGY

The NourishShare algorithm is intended to match excess food to nearby NGOs by using distance and time constraints. Firstly, nearby NGOs situated in a defined distance from the donor organization are chosen as suitable matches for the donations. In addition, these NGOs are arranged according to their proximity to the donor, ensuring minimal travel time to deliver the donation. Secondly, donations are prioritized based on their expiry date, as perishable products should be delivered immediately. The matching algorithm will take into account distance and expiry time constraints, resulting in an effective allocation of excess food. The proposed algorithm relies on a scoring mechanism that takes distance and time expiry into consideration when matching donors to NGOs. Thus, the score of matching is calculated as follows:

$$\text{Score} = (w1 \times \text{Distance}) + (w2 \times \text{Expiry Urgency})$$

where $w1$ and $w2$ represent weights associated with each parameter. Thus, a smaller score reflects a more desirable match.

Such a scoring mechanism will minimize transportation time and prevent food decay.

3.6 DATABASE SCHEMA DESIGN

The database design ensures efficient data access and scalability.

Collection	Fields	Description
Users	UserID, Name, Email, Role	Stores donor and NGO details
FoodPosts	PostID, DonorID, FoodType, Quantity, Latitude, Longitude, ExpiryTime	Stores donation information
NGO Claims	ClaimID, NGOID, PostID, ClaimTimestamp, Status	Tracks donation claims

CHAPTER 4:

IMPLEMENTATION AND RESULT

4.1 IMPLEMENTATION OVERVIEW

The NourishShare system is developed to use a full-stack web architecture to facilitate real-time communication between food donors and NGOs. It uses the MERN stack, which includes MongoDB (database), Express.js and Node.js (backend), and React.js (frontend). The architecture adopts a modular development strategy, with individual components for user interaction, processing, databases, and geospatial. These interact through RESTful APIs for scalability, maintainability, and seamless data exchange. The system emphasizes features such as real-time posting of food donations, geospatial matching of donors and NGOs, and user authentication with claim processing.

4.2 FRONTEND IMPLEMENTATION

The NourishShare system's user interface is built with React.js and Tailwind CSS for a responsive and mobile-friendly design. The system includes role-based dashboards with specific features for donors and NGOs. Donors can post and manage food donations, and NGOs can access local donations using a map view. Real-time mapping of food donation locations is made possible by the integration of Google Maps, which enhances user experience and decision-making. Axios facilitates effective data transport by facilitating API calls between the frontend and backend applications.

4.3 BACKEND IMPLEMENTATION

Node.js and Express.js are used in the system's backend to manage API calls and business logic. Important tasks include user authentication using JSON Web Tokens, posting and retrieving food donations, coordinate processing (latitude and longitude), distance filtering to locate the closest NGOs, and donation claims are all handled by the backend. All requests are verified by middleware to make sure they are legitimate and include accurate data. System

functionality that is safe, quick, and scalable is ensured by this backend design.

4.4 DATABASE IMPLEMENTATION

MongoDB Atlas, a cloud-based NoSQL database, is used by the NourishShare system to store and retrieve data. Users (donors and NGOs), Food Posts (which include information about food kind, quantity, location, and expiration time), and NGO Claims (which include information about donor claims and their statuses) make up the database's three primary collections. The MongoDB 2dsphere index is used to index the location fields, enabling effective spatial queries and real-time matching.

4.5 API DESIGN

The platform connects the frontend and backend via a number of RESTful API endpoints. These APIs make it easier to administer food donations, handle user verification, and process claims in a safe and organized manner. For consistency and integration, endpoints follow HTTP methods (such as GET and POST).

Endpoint	Method	Description
/api/auth/register	POST	User registration
/api/auth/login	POST	User authentication
/api/posts/create	POST	Create a food donation.
/api/posts/list	GET	Retrieve food posts
/api/posts/claim	POST	NGO claims donation

The interchange of structured data between platforms is verified by these APIs.

4.6 EXPERIMENTAL SETUP

Several users who represented food donors and non-governmental organizations tested the system in a controlled setting. A variety of scenarios were carried out, including taking donations, publishing food donations, showing information about available donations on the map, and spatially selecting NGOs according to distance. These tests were carried out to assess the system's coordination, efficacy, and efficiency in practical use cases.

4.6 PERFORMANCE EVALUATION

API response time, matching efficiency, and system scalability were used to gauge the system's performance. Less than 200 milliseconds was the average response time for the API call, indicating effective processing. The matching efficiency was improved by the distance filtering feature, which significantly reduced the search time for NGOs. Additionally, the system's scalability was demonstrated by its capacity to service several customers with little delay. These results imply that real-time food redistribution efforts could be facilitated by the system.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

Food waste is still a major problem even with an abundance of food supplies. Current food donation programs frequently lack efficient coordination and prompt communication between NGOs and food donors.

Geospatial matching and web technologies are combined in the NourishShare system to provide an online real-time solution. The method efficiently integrates the Google Maps API with the MERN stack to help NGOs and food donors coordinate.

In order to minimize redirection delays and improve redistribution, the system guarantees effective matching based on geographic proximity and user authentication. Low-latency responses and real-time operations are supported by the system's performance.

In conclusion, the strategy offers a technology-based, scalable, and workable solution to urban food redistribution and has the potential to be implemented and have a beneficial social impact.

5.2 FUTURE SCOPE

Although the NourishShare system's current implementation provides a workable method for food redistribution, there are a few enhancements that could make the system even better. Proactive resource management can be facilitated by using machine learning algorithms in conjunction with predictive analytics to anticipate the demands of NGOs and surplus food production. Additionally, using optimization techniques in food collecting might save trip time and boost productivity.

A mobile app would facilitate on-the-ground operations and increase accessibility. Additionally, NGOs can be informed as soon as food contributions become available in the area by using real-time notifications. Understanding the system's scalability and efficacy can also be improved by large-scale deployment and implementation using actual data. The platform may become a more intelligent, effective, and accessible food redistribution system as a result of these developments.

REFERENCES

- [1] Food and Agriculture Organization of the United Nations (FAO), Global Food Losses and Food Waste: Extent, Causes, and Prevention. Rome, Italy: FAO, 2011.
- [2] United Nations Environment Programme (UNEP), Food Waste Index Report 2021. Nairobi, Kenya: UNEP, 2021.
- [3] S. Shekhar and H. Xiong, Encyclopedia of GIS, 2nd ed. New York, NY, USA: Springer, 2017.
- [4] J. H. Kim and M. S. Park, "Geospatial indexing techniques for real-time location-based services," IEEE Access, vol. 8, pp. 145203–145214, 2020.
- [5] R. G. Ballou, Business Logistics and Supply Chain Management, 5th ed. Upper Saddle River, NJ, USA: Pearson, 2004.
- [6] M. Dorigo and T. Stützle, Ant Colony Optimization. Cambridge, MA, USA: MIT Press, 2004.
- [7] G. Laporte, "The vehicle routing problem: An overview of exact and approximate algorithms," European Journal of Operational Research, vol. 59, no. 3, pp. 345–358, 1992.
- [8] A. K. Sangaiah et al., "Internet of Things and big data analytics for smart and connected communities," Future Generation Computer Systems, vol. 76, pp. 312–326, 2017.
- [9] R. Buyya, C. S. Yeo, and S. Venugopal, "Market-oriented cloud computing: Vision, hype, and reality for delivering IT services as computing utilities," Future Generation Computer Systems, vol. 25, no. 6, pp. 599–616, 2009.
- [10] M. Stonebraker and U. Çetintemel, "One size fits all: An idea whose time has come and gone," in Proceedings of the IEEE International Conference on Data Engineering, 2005.

- [11] R. W. Sinnott, "Virtues of the Haversine," *Sky and Telescope*, vol. 68, no. 2, pp. 159–160, 1984.
- [12] A. Verma and S. Kumar, "Time-sensitive resource allocation in perishable supply chains," *International Journal of Production Economics*, vol. 230, 2020.
- [13] K. Chodorow, *MongoDB: The Definitive Guide*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.
- [14] L. Richardson and S. Ruby, *RESTful Web Services*. Sebastopol, CA, USA: O'Reilly Media, 2007.
- [15] M. Kleppmann, *Designing Data-Intensive Applications*. Sebastopol, CA, USA: O'Reilly Media, 2017.
- [16] A. Silberschatz, H. F. Korth, and S. Sudarshan, *Database System Concepts*, 7th ed. New York, NY, USA: McGraw-Hill, 2019.
- [17] R. Dhanaraj, S. K. Hafizul Islam, and V. Rajasekar, "A cryptographic paradigm to detect and mitigate blackhole attack in VANET environments," *Wireless Networks*, vol. 28, no. 7, pp. 3127–3142, 2022.
- [18] S. Dhiviya, S. Malathy, and D. Rajesh Kumar, "Internet of Things (IoT) elements, trends, and applications," *Journal of Computational and Theoretical Nanoscience*, vol. 15, no. 5, pp. 1639–1643, 2018.
- [19] M. A. Ali, R. K. Dhanaraj, and A. Nayyar, "A high-performance-oriented AI-enabled IoT-based pest detection system using sound analytics in large agricultural fields," *Microprocessors and Microsystems*, vol. 103, p. 104946, 2023.
- [20] P. Kherwa et al., "Contextual embedded text summarizer system: A hybrid approach," *Expert Systems*, vol. 42, no. 2, p. e13733, 2025.
- [21] S. Kaur et al., "High-accuracy lung disease classification via logistic regression and advanced feature extraction techniques," *Egyptian Informatics Journal*, vol. 29, p. 100596, 2025.

[22] R. Kaur et al., "Development of a cryptocurrency price prediction model: Leveraging GRU and LSTM for Bitcoin, Litecoin, and Ethereum," *PeerJ Computer Science*, vol. 11, p. e2675, 2025.

APPENDIX 1

APPENDIX A: SYSTEM ARCHITECTURE DIAGRAM

The NourishShare platform's high-level architecture shows how the database system, backend services, frontend interface, and external geolocation APIs interact. In order to provide effective data sharing and real-time updates, the system uses a modular full-stack architecture in which the frontend and backend interact via RESTful API calls. Google Maps services, which facilitate location-based features like geocoding and map rendering, are integrated with geospatial processing. The architecture diagram also shows how the Node.js backend, MongoDB database, React-based frontend, and Google Maps API work together to provide smooth coordination for real-time food donation and redistribution.

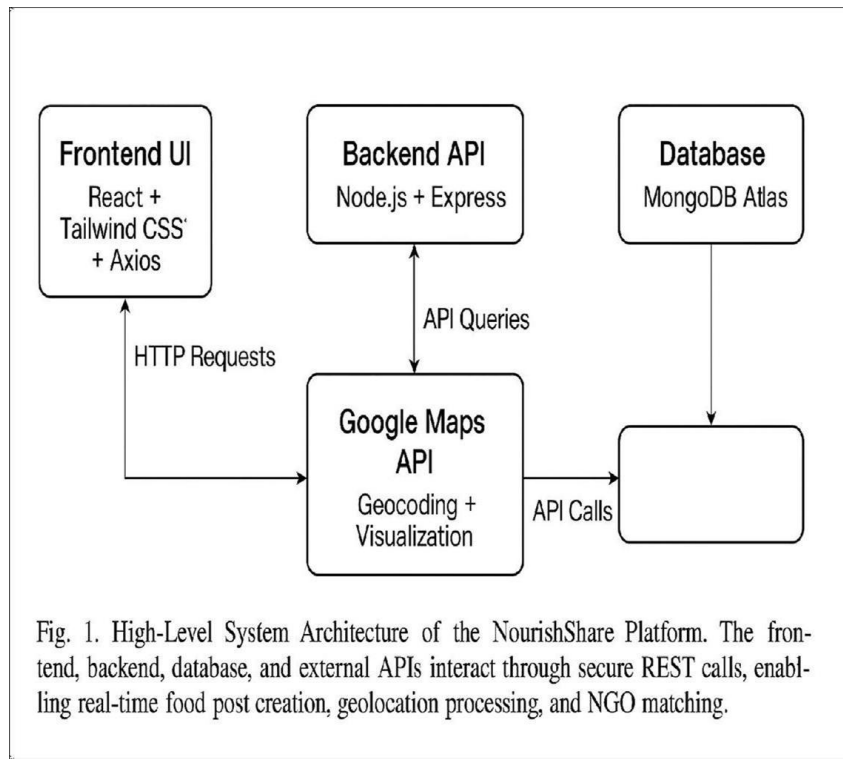


Figure A1: High-Level System Architecture of the NourishShare Platform

APPENDIX B: KEY SCREEN DESIGNS

This section outlines the main user interface designs in NourishShare and shows the user experience for both donors and NGOs. The home interface is the initial screen that gives users access to the main functions of the system and explains the aim of the system to reduce food waste by linking food donations to nearby NGOs. The map interface for food donations shows the locations of all active donations on a map and allows NGOs to easily select food donations for collection based on proximity.

The donor interface for food submission provides a way for users to input information about food donations, such as the food type, amount, location, and food expiry time. The mobile interface examples also showcase the responsiveness and ease of use of the system and the usefulness of the donation submission and real-time map visualization features in improving the user experience. The mobile interface examples in the system illustrate the donation interface and the live map.



Figure B1: NourishShare Mobile Interface

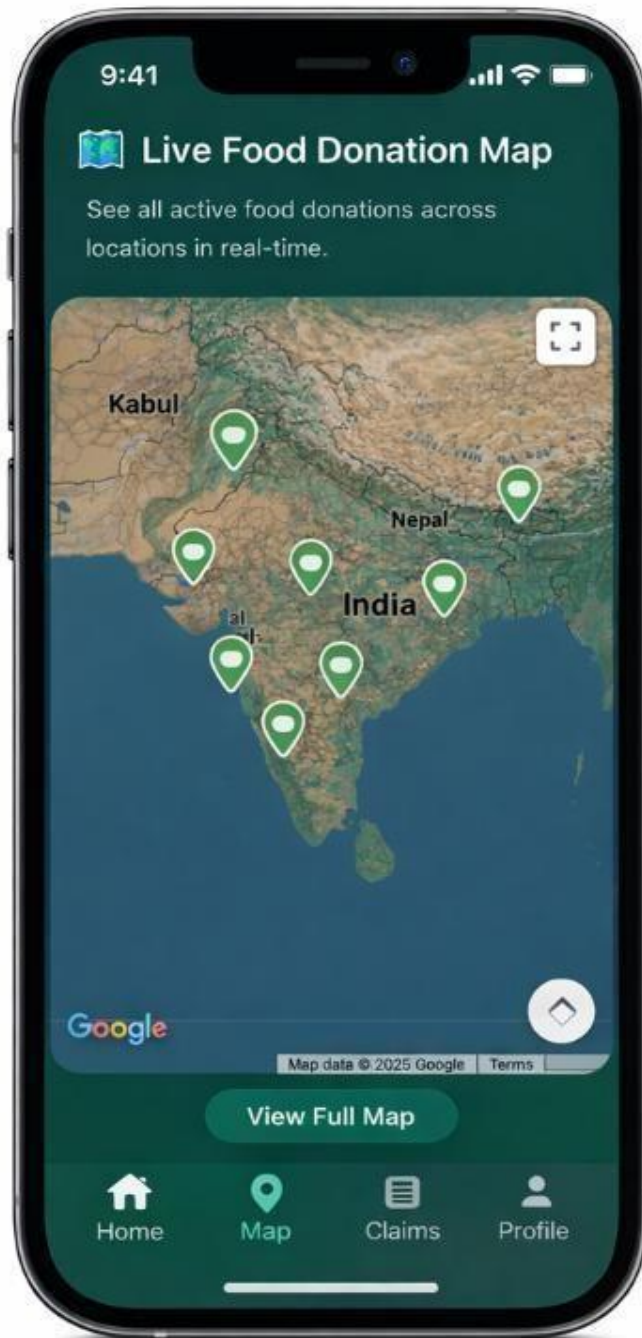


Figure B2: Live Food Donation Map Visualization

APPENDIX C: API ENDPOINTS

The front-end application can communicate with the backend services through the RESTful API that the NourishShare system offers. These endpoints are in charge of safely handling crucial functions including NGO claims, food contribution processing, and user identification. For stability and dependability, popular HTTP techniques like GET and POST are used. The system as a whole is enhanced by the API structure's ability to facilitate seamless data exchange and real-time coordination between donors and NGOs.

Endpoint	Method	Description
/api/auth/register	POST	Registers a new user
/api/auth/login	POST	Authenticates user credentials
/api/posts/create	POST	Creates a new food donation post
/api/posts/list	GET	Retrieves all available food posts
/api/posts/claim	POST	Allows NGOs to claim a donation

The frontend and backend components can communicate securely thanks to these endpoints.

APPENDIX C: DATABASE SCHEMA

The NourishShare database schema is designed with real-time geospatial processing and data storage and retrieval in mind. Users, Food Posts, and NGO Claims are the three primary collections in the system. Basic user data, including user ID, name, email address, role (donor or NGO), and contact details, are included in the Users collection. Information on food donations, such as post ID, donor ID, food type, quantity, location (latitude and longitude), and expiration date, is included in the Food Posts collection.

Donation claim information, including claim ID, NGO ID, post ID, claim time, and claim status, are stored in the NGO claims collection. Food donations can be managed efficiently thanks to this template. Additionally, location data employ geographical indexing to facilitate spatial queries, guaranteeing effective identification of neighboring NGOs. We can track food donations until they are claimed thanks to this schema, which also facilitates effective geolocation search.

PATENT

PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION

PLAGIARISM REPORT